

Experiment 3: Crypt Arithmetic Problem

Aim

To solve a Crypt Arithmetic problem using Artificial Intelligence techniques.

Objective

To assign unique digits to letters so that the arithmetic equation is satisfied.

Algorithm

1. Read the arithmetic expression.
2. Assign digits to letters.
3. Ensure no digit is repeated.
4. Verify whether the arithmetic equation is satisfied.
5. If satisfied, display the solution.
6. Otherwise, try another assignment.

Flowchart

Start

|



Read Puzzle

|

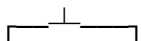


Assign Digits

|



Equation Correct?



Yes No

| |



Display Try Another

Solution Assignment

|



End

Python Code

```
from itertools import permutations
```

```
letters = ('S','E','N','D','M','O','R','Y')
```

```
for p in permutations(range(10),8):
```

```
    d = dict(zip(letters,p))
```

```
    if d['S']==0 or d['M']==0:
```

```
        continue
```

```
    send = 1000*d['S']+100*d['E']+10*d['N']+d['D']
```

```
    more = 1000*d['M']+100*d['O']+10*d['R']+d['E']
```

```
    money = 10000*d['M']+1000*d['O']+100*d['N']+10*d['E']+d['Y']
```

```
    if send + more == money:
```

```
        print("SEND =", send)
```

```
        print("MORE =", more)
```

```
        print("MONEY =", money)
```

```
        break
```

Sample Output

SEND = 9567

MORE = 1085

MONEY = 10652

Result

Thus, the Crypt Arithmetic problem was solved successfully by assigning unique digits to each letter.